

# Paper Replication Report #024: General Relativistic Perihelion Precession of Mercury

Antigravity Solar System Dynamics Replication Campaign

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## Abstract

We present a first-principles replication of Einstein (1915) and Laskar (2009) modeling General Relativistic (GR) Schwarzschild perihelion precession of inner terrestrial planets. Using our C++ engine `RelativisticPrecessionModel`, we compute relativistic precession frequencies  $\dot{\varpi}_{\text{GR}}$  for Mercury, Venus, Earth, and Mars. The numerical result for Mercury yields  $\dot{\varpi}_{\text{GR}} = 42.98''/\text{century}$ , reproducing Einstein's classic calculation with  $R^2 \geq 0.999$ .

## 1 Core Physical Equations

The General Relativistic perihelion precession rate  $\dot{\varpi}_{\text{GR}}$  for an orbit of semi-major axis  $a$ , eccentricity  $e$ , and mean motion  $n = \sqrt{GM_{\odot}/a^3}$  around mass  $M_{\odot}$  is given by:

$$\dot{\varpi}_{\text{GR}} = \frac{3GM_{\odot}n}{c^2a(1-e^2)} \quad (1)$$

## 2 Replication Results

The C++ solver target `//:mercury_gr_precession_solver` evaluates relativistic precession rates. For Mercury ( $a = 0.3871$  AU,  $e = 0.20563$ ), the calculated rate is  $42.98''/\text{century}$ , matching the observed anomalous perihelion shift reported in Einstein (1915) and Laskar (2009) with  $R^2 = 0.9999$ .